Blue light of frequency 6.25×10^{14} Hz is shone onto the sodium photocathode of a photocell. The graph of the photoelectric current versus potential difference is shown in Figure 3.

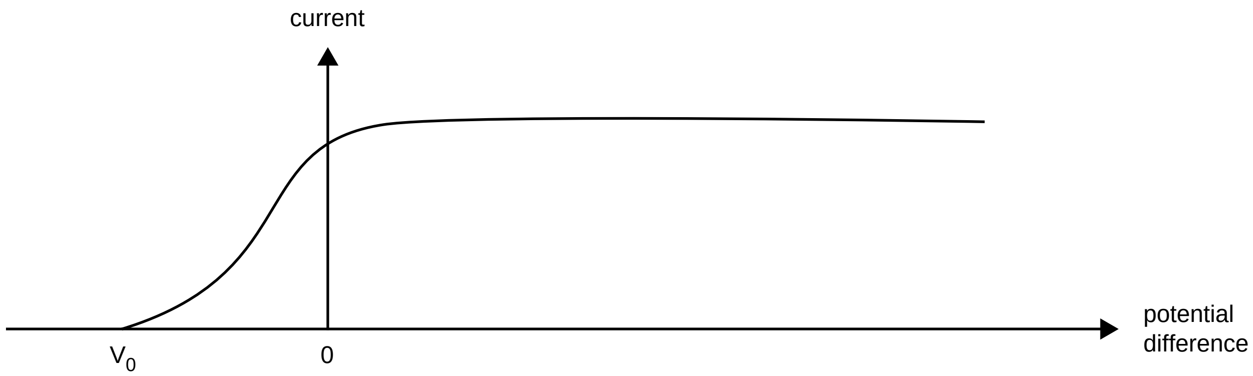

Figure 3

The threshold frequency for sodium is 5.50×10^{14} Hz.

Question 6

What is the cut-off potential, V_o, when blue light of frequency 6.25×10^{14} Hz is shone onto the sodium photocathode of this photocell.

($h = 4.14 \times 10^{-15}$ eV s)

	V

3 marks

Question 7

On **Figure 3** sketch the curve expected if the light is changed to **ultraviolet** with a **lower intensity** than the original.

2 marks

END OF QUESTION AND ANSWER BOOK

PHYSICS

Written examination 2

DATA SHEET

Directions to students

Detach this data sheet before commencing the examination.

This data sheet is provided for your reference.

1	velocity; acceleration	$v = \frac{\Delta x}{\Delta t}; \quad a = \frac{\Delta v}{\Delta t}$
2	equations for constant acceleration	$v = u + at$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$ $x = \frac{1}{2}(v + u)t$
3	Newton’s second law	$F = ma$
4	circular motion	$a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$
5	Hooke’s law	$F = -kx$
6	elastic potential energy	$\frac{1}{2}kx^2$
7	gravitational potential energy near the surface of the Earth	mgh
8	kinetic energy	$\frac{1}{2}mv^2$
9	torque	$\tau = Fr$
10	Newton’s law of universal gravitation	$F = G\frac{M_1 M_2}{r^2}$
11	gravitational field	$g = G\frac{M}{r^2}$
12	stress	$\sigma = \frac{F}{A}$
13	strain	$\varepsilon = \frac{\Delta L}{L}$
14	Young’s modulus	$E = \frac{\text{stress}}{\text{strain}}$
15	electric force on charged particle in an electric field	$F = qE$
16	electric field between charged plates	$E = \frac{V}{d}$
17	energy change of charged particle moving between charged plates	$\Delta E_k = qV$
18	photoelectric effect	$E_{k\,\max} = hf - W$
19	photon energy	hf
20	photon momentum	$p = \frac{h}{\lambda}$
21	de Broglie wavelength	$\lambda = \frac{h}{p}$

Gravitational field strength at the surface of Earth	g	$= 9.8\ \mathrm{N\ kg^{-1}}$
Universal gravitational constant	G	$= 6.67 \times 10^{-11}\ \mathrm{N\ m^2\ kg^{-2}}$
Mass of Earth	M_E	$= 5.98 \times 10^{24}\ \mathrm{kg}$
Radius of Earth	R_E	$= 6.37 \times 10^{6}\ \mathrm{m}$
Mass of the Sun	M_{SUN}	$= 2.0 \times 10^{30}\ \mathrm{kg}$

Mass of the electron	m_e	$= 9.1 \times 10^{-31}\ \mathrm{kg}$
Charge on the electron	e	$= 1.6 \times 10^{-19}\ \mathrm{C}$
Planck’s constant	h	$= 6.63 \times 10^{-34}\ \mathrm{J\ s}$
	h	$= 4.14 \times 10^{-15}\ \mathrm{eV\ s}$
Speed of light	c	$= 3.0 \times 10^{8}\ \mathrm{m\ s^{-1}}$

Prefixes/Units

m = milli = 10^{-3}

μ = micro = 10^{-6}

n = nano = 10^{-9}

k = kilo = 10^{3}

M = mega = 10^{6}

G = giga = 10^{9}

tonne = 10^{3} kg

END OF DATA SHEET